

Using the data health systems already collect to make high-stakes care safer and more efficient.

WHO WE ARE

Afferent Health LLC turns the data a health system already collects into decisions clinicians can act on and defend. We learn from the whole record - waveforms, monitor streams, laboratory results, longitudinal EHR data, and clinical text - and translate it into accountable decision-support for health systems, federal partners, and technology teams.

We work where the data are richest and the stakes are highest: surgery, intensive care, and other complex clinical workflows. We choose the architecture that performs best, then earn trust through calibration, physiologic checks, subgroup and transfer testing, negative controls, disclosed limits, and human-use design.

CORE CAPABILITIES

- **Complete-record clinical intelligence.** Cohort definition, record linkage, governance, and analysis across waveforms, monitor numerics, labs, EHR data, clinical text, and longitudinal history.
- **Multimodal waveform and signal processing.** Learning from raw arterial pressure, capnography, pulse oximetry, and EEG morphology - not only the summary numbers a monitor reports.
- **Accountable ML/AI and clinical TEVV.** Prototype development, calibration, subgroup and site-transfer testing, explainability, abstention, safety analysis, negative controls, drift monitoring, and test/evaluation/verification/validation support.
- **Perioperative, ICU, and acute-care intelligence.** Risk stratification and monitoring for surgical outcomes, biological age and reserve, delirium, deterioration, kidney injury, respiratory events, readmission, and care transitions.
- **Clinical workflow and decision-support design.** Turning clinical requirements into usable tools for transplant readiness, clinical summaries, release gates, surveillance, OR flow, coding review, and care navigation.
- **Research-to-product and value translation.** Evidence synthesis, data-feasibility review, study design, proof-of-concept planning, prospective validation, and auditable economic-impact and budget models.
- **Federal health solution shaping.** Clinical requirements translation, RFI/RFP and innovation-funding responses, bid/no-bid analysis, work-package design, vehicle-fit analysis, teaming, and specialized clinical or AI/ML personnel.

DIFFERENTIATORS

- **Physician-scientist depth.** Clinical validity, workflow fit, and measurable impact are designed in from the start by clinicians and scientists who understand both the signal and the decision it must support.
- **Existing data first.** When a health system already captures the signal, we make that record more useful before asking for new hardware or unnecessary workflow.

PARTNER ECOSYSTEM



COMPANY DATA

LEGAL ENTITY	PRIMARY NAICS	ADDITIONAL NAICS	CONTACT
Afferent Health LLC New York, NY Small business	541690 Other Scientific and Technical Consulting Services	541715 - Research and Development in the Physical, Engineering, and Life Sciences (except Nanotechnology and Biotechnology) 541512 - Computer Systems Design Services	afferenthealth.com dkaw.ventures@gmail.com

Afferent leads clinical and scientific intelligence; partners extend engineering, infrastructure, integration, and federal delivery.

KEY PERSONNEL

- **D. K. Adrian Williams, MSc - Founder & Managing Principal.** Physician-scientist in training, clinician-inventor, and clinical-modeling lead working in multimodal physiology, pulse oximetry, capnography, EEG, and research-to-product translation.
- **Joseph R. Scarpa, MD-PhD - Medical Consultant.** Anesthesiologist and NIA-funded investigator in perioperative machine learning, biological aging, surgical outcomes, and clinical workflow design.
- **Ruben Coen-Cagli, PhD - Scientific/AI Consultant.** Machine-learning methods investigator focused on high-dimensional signals, probabilistic modeling, computational neuroscience, and rigorous model development.

CURRENT WORK AND PRODUCT DIRECTIONS

Afferent is a newly formed firm. Our experience is scientific, clinical, and product-development work rather than prior prime-contract performance.

- **Surgical Intelligence.** Biological age and reserve, delirium prevention, perioperative risk, failure-to-rescue and readmission, ICU escalation, and OR readiness or flow.
- **Transplant decision support.** LLM-assisted referral readiness, packet quality, clinical navigation, and review support designed around documented transplant-service friction.
- **Clinical AI assurance.** TEVV and release gates for calibration, bias, missingness, drift, cross-site transfer, explanation, human review, and audit-ready evidence.
- **Physiologic intelligence.** Pulse-oximetry equity, capnography, arterial-waveform, EEG, and multimodal decompensation methods built from signals already present in routine care.
- **Clinical data and operations.** Clinical-summary quality, coding and appeals support, opioid-safety surveillance, workforce and rotation operations, and data-trust or reconciliation workflows.
- **Federal capture intelligence.** Structured opportunity monitoring, contract-vehicle fit, clinical work-package design, and evidence-backed concepts for pilots, task orders, grants, and R&D programs.